

Prolog Program to Implement Monkey Banana Problem

Aim

To write a Prolog program to implement the **Monkey Banana Problem** using states and actions.

Algorithm

1. Start with the monkey on the floor.
2. Move the monkey to the box.
3. Push the box below the banana.
4. Climb onto the box.
5. Take the banana.
6. Display that the monkey gets the banana.
7. Stop.

Prolog Program

```
% Monkey Banana Problem

state(at_door, on_floor, at_window, has_not).

solve :-
    state(at_door, on_floor, at_window, has_not),
    write('Monkey moves to window'), nl,

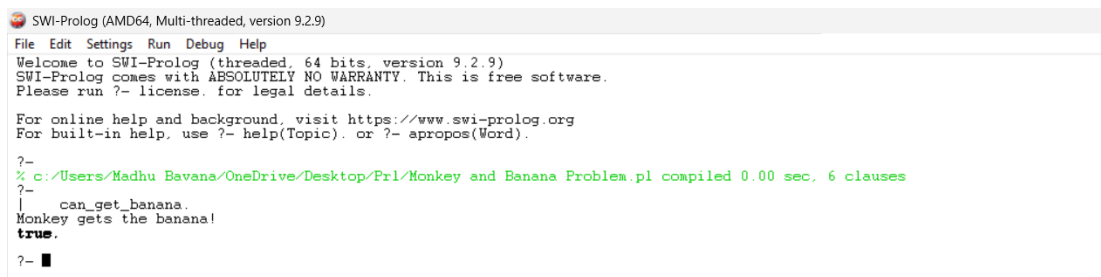
    write('Monkey pushes box under banana'), nl,

    write('Monkey climbs on box'), nl,

    write('Monkey takes banana'), nl,

    write('Monkey gets the banana!').
```

OUTPUT

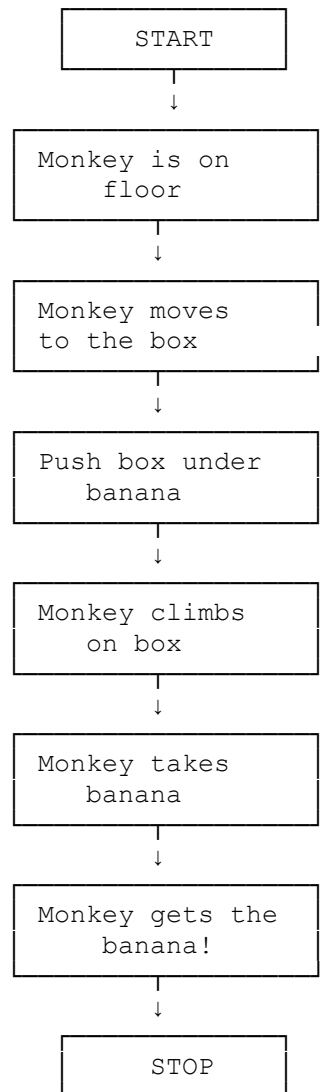


```
SWI-Prolog (AMD64, Multi-threaded, version 9.2.9)
File Edit Settings Run Debug Help
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?-
% c:\Users\Madhu Bavana\OneDrive\Desktop\Pr1\Monkey and Banana Problem.pl compiled 0.00 sec, 6 clauses
?-
| can_get_banana.
Monkey gets the banana!
true.
?-
```

Flowchart – Monkey Banana Problem



Conclusion

Thus, the **Monkey Banana Problem** is successfully implemented in Prolog, and the monkey obtains the banana by performing a sequence of actions.